

AI-Powered Medical Prescription Intelligence System Combined Architecture & MongoDB Schema

1. System Overview

An end-to-end AI system that converts prescription images into structured medical intelligence including entity extraction, drug interaction reasoning, risk scoring, and explainable reporting. The system follows a perception → understanding → reasoning → reporting paradigm.

2. End-to-End Pipeline

- User Authentication and Role-based Access
- Image Upload and Storage
- Image Preprocessing and Quality Assessment
- OCR Perception Layer
- Entity Extraction and Structuring
- Drug Name Normalization
- Drug Interaction Reasoning
- Prescription Risk Scoring
- Knowledge Enrichment and Reporting
- Dashboard Analytics and Event Logging

3. Final Tech Stack

- Frontend: React / Next.js, TailwindCSS, Chart.js, Socket.io
- Backend: Node.js + Express modular architecture
- AI Services: Python microservices (YOLOv8, OpenCV, PaddleOCR)
- Database: MongoDB (flexible AI schema)
- Storage: Local/S3 for images and reports
- Cache & Realtime: Redis + Socket.io
- Infrastructure: Docker Compose deployment

4. Architectural Style

- Modular Monolith with clearly separated service folders

- Node backend orchestrates AI microservices
- Tunable configuration and versioned knowledge base
- Explainability via provenance storage
- Event-driven analytics instrumentation

5. MongoDB Collection Design

- users
- prescriptions
- image_quality
- ocr_outputs
- entities
- drug_synonyms
- interaction_kb
- interaction_results
- risk_scores
- risk_reasons
- events
- annotations
- config_versions

6. Core Collection Descriptions

- Users: authentication and role management
- Prescriptions: central pipeline object
- Image Quality: preprocessing metrics and crop data
- OCR Outputs: token and confidence storage
- Entities: structured extracted prescription elements
- Drug Synonyms: normalization mapping
- Interaction KB: curated drug interaction knowledge
- Interaction Results: detected interactions per prescription
- Risk Scores: aggregated multi-signal risk
- Risk Reasons: explainability factors
- Events: analytics event logging
- Annotations: dataset bounding boxes
- Config Versions: tunable configuration storage

7. Design Principles

- Explainable AI with evidence preservation
- Tunable and versioned decision parameters
- Separation of perception, understanding, reasoning, and presentation
- Analytics-first instrumentation
- Hackathon feasibility with production-aligned thinking

8. Jury Value Proposition

- End-to-end explainable medical AI pipeline
- Hybrid perception and knowledge reasoning architecture
- Configurable risk scoring framework
- Curated clinical knowledge base integration
- Real-time analytics and human-in-the-loop capability